Modifiable Lifestyle Factors

Foundational Roots of the Functional Medicine Model

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APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

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Disclosures

Kelechi A. Uduhiri, MD disclosed no relevant financial relationships with any ineligible companies.

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Functional Medicine Certification Competencies

THERAPEUTIC INTERVENTION

a) Evaluate patient readiness to engage in dietary, lifestyle, and behavioral changes prior to initiating therapeutic interventions; prioritize these as initial interventional modalities.

b) According to their ability to favorably modulate function, correct clinical imbalances, and/or address each patient's personalized root causes and in alignment with the patient's goals, values, preferences, sociocultural context, and what gives them meaning and purpose:

   i. Identify, prioritize, select, and initiate the following therapeutic interventions when clinically indicated:
      1. Dietary and other lifestyle strategies.
      2. Behavioral change strategies.
      3. Pharmaceuticals, nutraceuticals, botanicals, and other supplements.
      4. Mind-body and other modalities.

   ii. Monitor changes in function in response to therapeutic interventions and adjust regimen accordingly.

   iii. Refer and/or collaborate with additional healthcare practitioners, as needed or required, to align with the patient's goals and desired outcomes.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

1. List the five modifiable lifestyle factors.

2. Explain the role of modifiable lifestyle factors as antecedents and mediators of physiological dysfunction and chronic disease.

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Clinical Guidelines

NIH – Lifestyle and Mental Health

- https://pubmed.ncbi.nlm.nih.gov/36202135/

Exercise and Lifestyle – Chronic Kidney Disease

- https://link.springer.com/article/10.1186/s12882-021-02618-1

AGA – Lifestyle for Weight Loss in MASLD

- https://www.sciencedirect.com/science/article/pii/S0016508520355384

AHA – Lifestyle and Cardiovascular Health (Pillars)

- https://www.ahajournals.org/doi/full/10.1161/CIR.0000000000001018

International Organizations x 3 – Lifestyle Management Of Hypertension

- https://journals.lww.com/jhypertension/fulltext/2024/01000/lifestyle_management_of_hypertension_.3.aspx

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Clinical Focus

- The scope of the problem of chronic disease.

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Non-Communicable Disease (NCDs)

- NCDs are the leading cause of death globally and have reached epidemic proportions.
- According to the World Health Organization (WHO), NCDs contributed to at least 43 million deaths in 2021; equivalent to 75% of non-pandemic-related deaths globally.
- The four major NCDs (cardiovascular disease, cancer, chronic respiratory diseases, and diabetes) accounted for 80% of all premature NCD-related deaths globally.

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World Health Organization. Noncommunicable diseases. Updated December 23, 2024. Accessed June 5, 2025. https://www.who.int/news-room/fact-sheets/detail/noncommunicable-diseases

Non-Communicable Disease (NCDs) and Modifiable Lifestyle Factors

- NCDs are diseases of lifestyle.
- NCDs are preventable.
- Changes in lifestyle behaviors have more potential for preventing and mitigating NCDs than any other type of intervention or public health policy.

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World Health Organization. Noncommunicable diseases. Updated December 23, 2024. Accessed June 5, 2025. https://www.who.int/news-room/fact-sheets/detail/noncommunicable-diseases

Chronic Diseases in the US

Financial and Human Scope

Of the annual $4.1 trillion healthcare expenditure in the US, 90% is attributed to managing and treating chronic diseases and mental health conditions.

In the US:

- Five out of the ten leading causes of death are strongly associated with preventable and treatable chronic diseases.
- 129 million people have at least one major chronic health condition.
- 42% have two or more chronic health conditions.
- 12% have at least five chronic health conditions.

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- National Center for Chronic Disease Prevention and Health Promotion. About chronic diseases. Centers for Disease Control and Prevention. Published October 4, 2024. Accessed May 28, 2025. https://www.cdc.gov/chronic-disease/about/index.html#cdc_disease_basics_overview-chronic-diseases-in-america
- Benavidez GA, Zahnd WE, Hung P, Eberth JM. Chronic disease prevalence in the US: sociodemographic and geographic variations by zip code tabulation area. *Prev Chronic Dis*. 2024;21:E14 doi:10.5888/pcd21.230267

The Chronic Disease Problem

$47 trillion globally by 2030

It is projected that the cost of chronic disease will be **$47 trillion globally by 2030.**

See "References: The Chronic Disease Problem"

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References

THE CHRONIC DISEASE PROBLEM

- Bloom DE, Cafiero ET, Jané-Llopis E, et al. *The Global Economic Burden of Non-communicable Diseases.* World Economic Forum; 2011. Accessed May 28, 2025. https://www3.weforum.org/docs/WEF_Harvard_HE_GlobalEconomicBurdenNonCommunicableDiseases_2011.pdf

- Bureau of Economic Analysis. Gross domestic product, third quarter 2024 (advance estimate). Published October 30, 2024. Accessed May 28, 2025. https://www.bea.gov/news/2024/gross-domestic-product-third-quarter-2024-advance-estimate

- Hacker K. The burden of chronic disease [published correction appears in *Mayo Clin Proc Innov Qual Outcomes*. 2024;9(1):100588. doi:10.1016/j.mayocpiqo.2024.11.005]. *Mayo Clin Proc Innov Qual Outcomes*. 2024;8(1):112-119. doi:10.1016/j.mayocpiqo.2023.08.005

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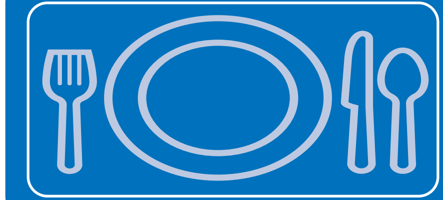
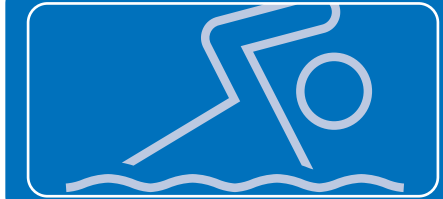
Chronic Disease

Social Determinants of Health as Antecedents and Mediators

Although individual health behaviors contribute to chronic diseases, they may not always be an individual choice but rather a circumstance due to socioeconomic and environmental factors.



Nutrition: Access to high quality, nutritious food may be limited for people with low income.



Physical Activity: Time limitations, financial constraints, and structural environmental factors (such as safe neighborhood, access to parks) can influence a person's ability to engage in regular physical activity.

Healthcare: Management of chronic disease requires the ability to afford healthcare and have physical access to it (i.e., access to transportation, pharmacies, hospitals). Research indicates that even when medically necessary, costly medications and healthcare can hinder one's access to it.

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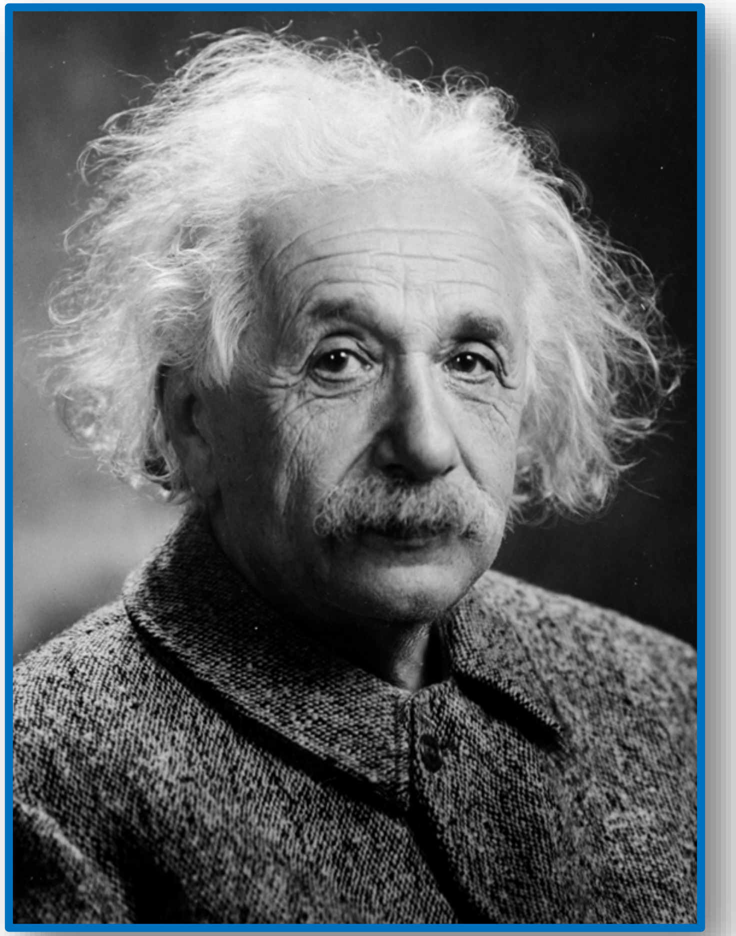
Benavidez GA, Zahnd WE, Hung P, Eberth JM. Chronic disease prevalence in the US: sociodemographic and geographic variations by zip code tabulation area. *Prev Chronic Dis*. 2024;21:E14. doi:10.5888/pcd21.230267

"A new type of thinking is essential if mankind is to survive and move toward higher levels."

\- Einstein



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Functional Medicine

Changing the way we do medicine
and the medicine we do.

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Clinical Focus

- The role of modifiable lifestyle factors as antecedents and mediators of chronic disease.

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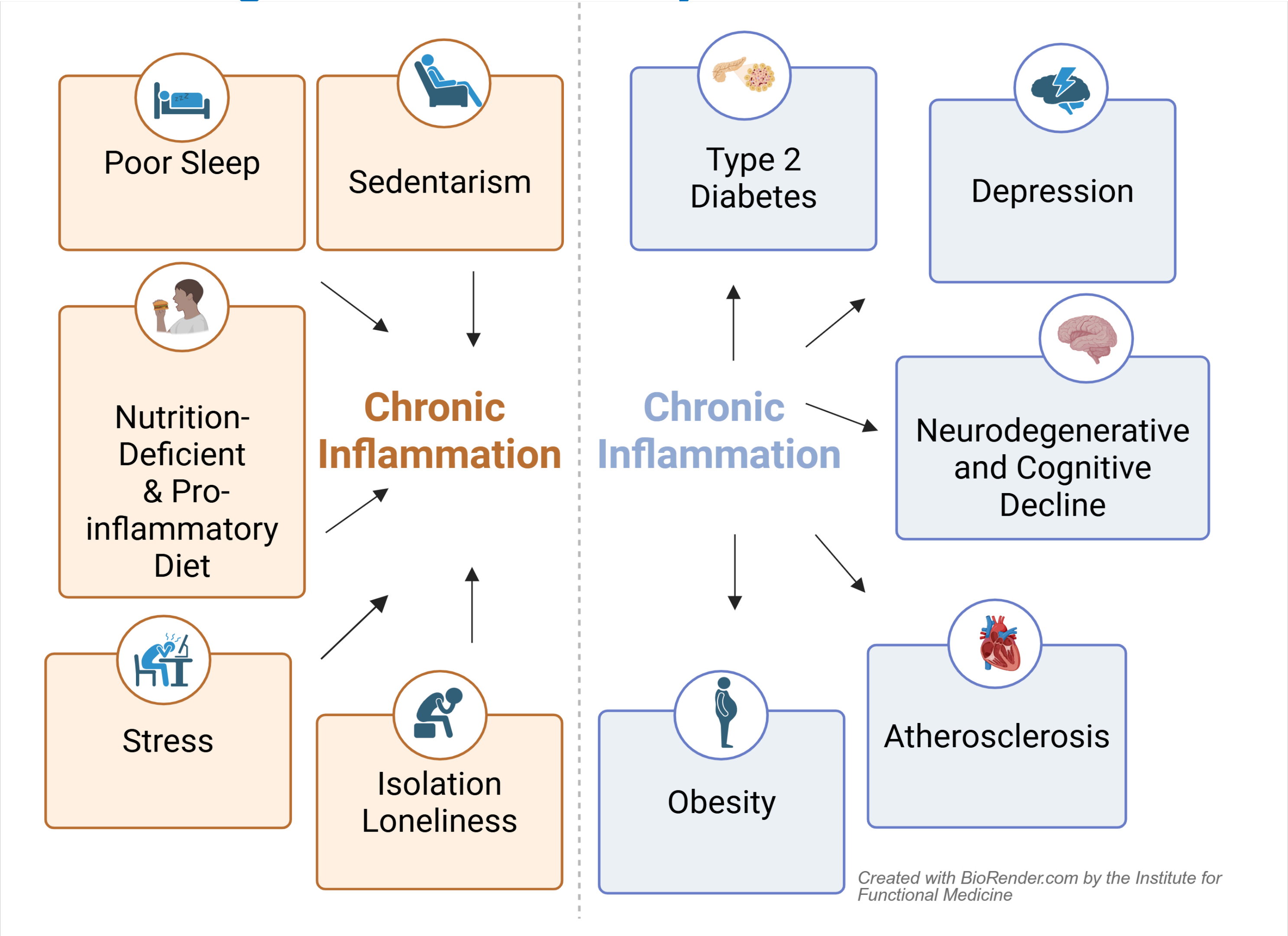
Chronic Inflammation

The Mechanistic Bridge Between Lifestyle Factors and Chronic Disease

One Condition
Many Imbalances



One Imbalance
Many Conditions

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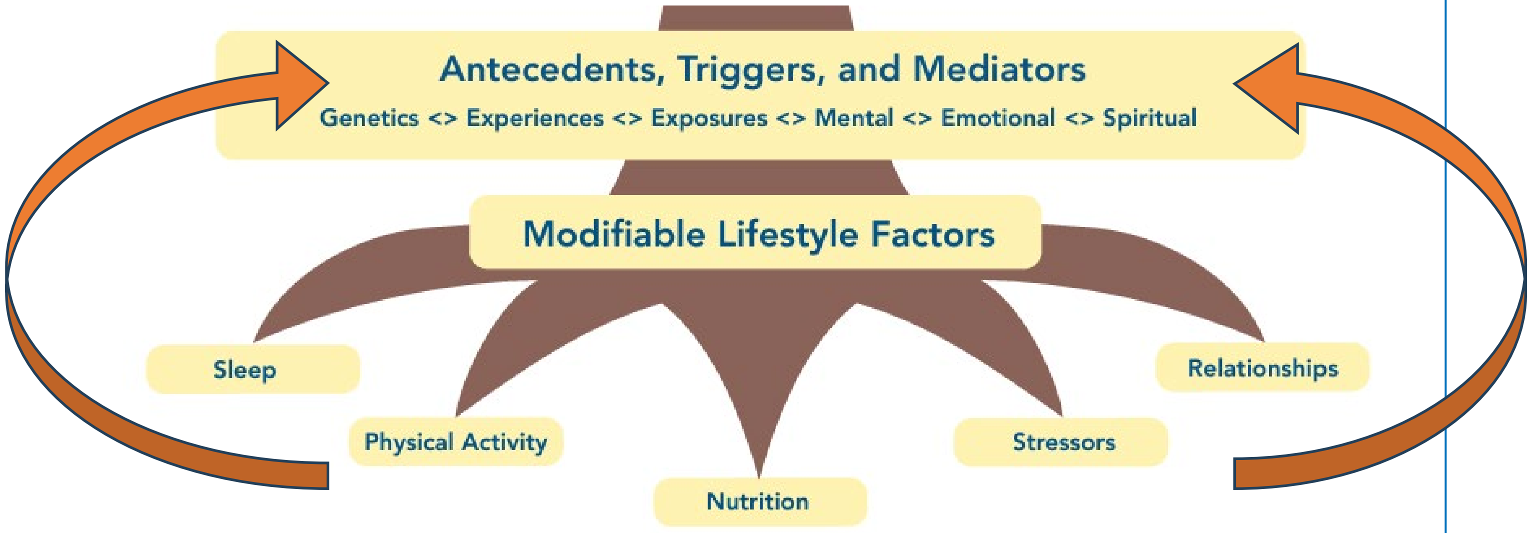
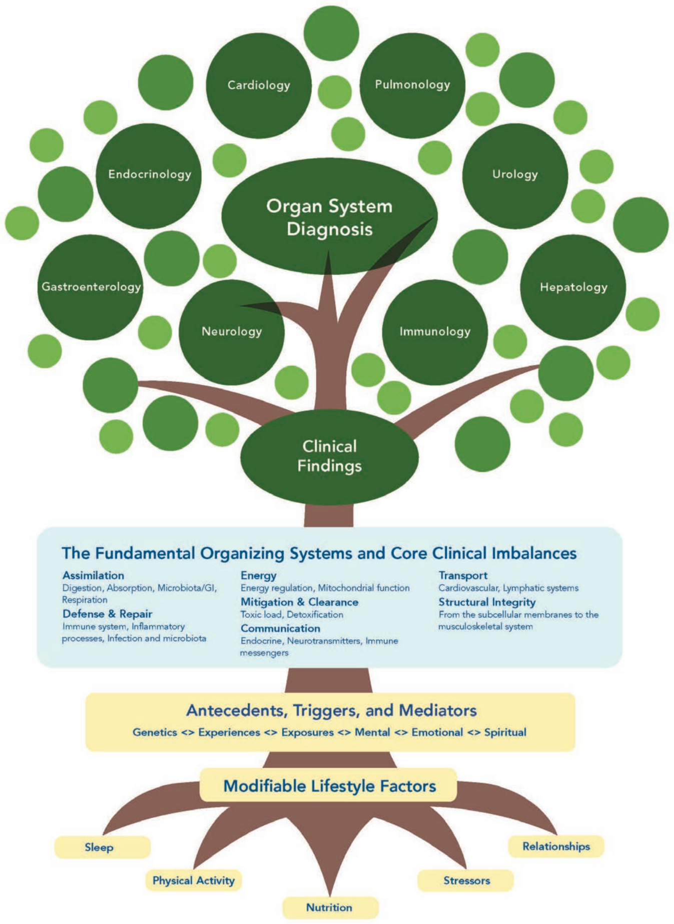
Lifestyle Factors as the Roots of the FM Model

The Role of MLFs in the Evaluation of the Patient

MLFs are considered antecedents and/or mediators.





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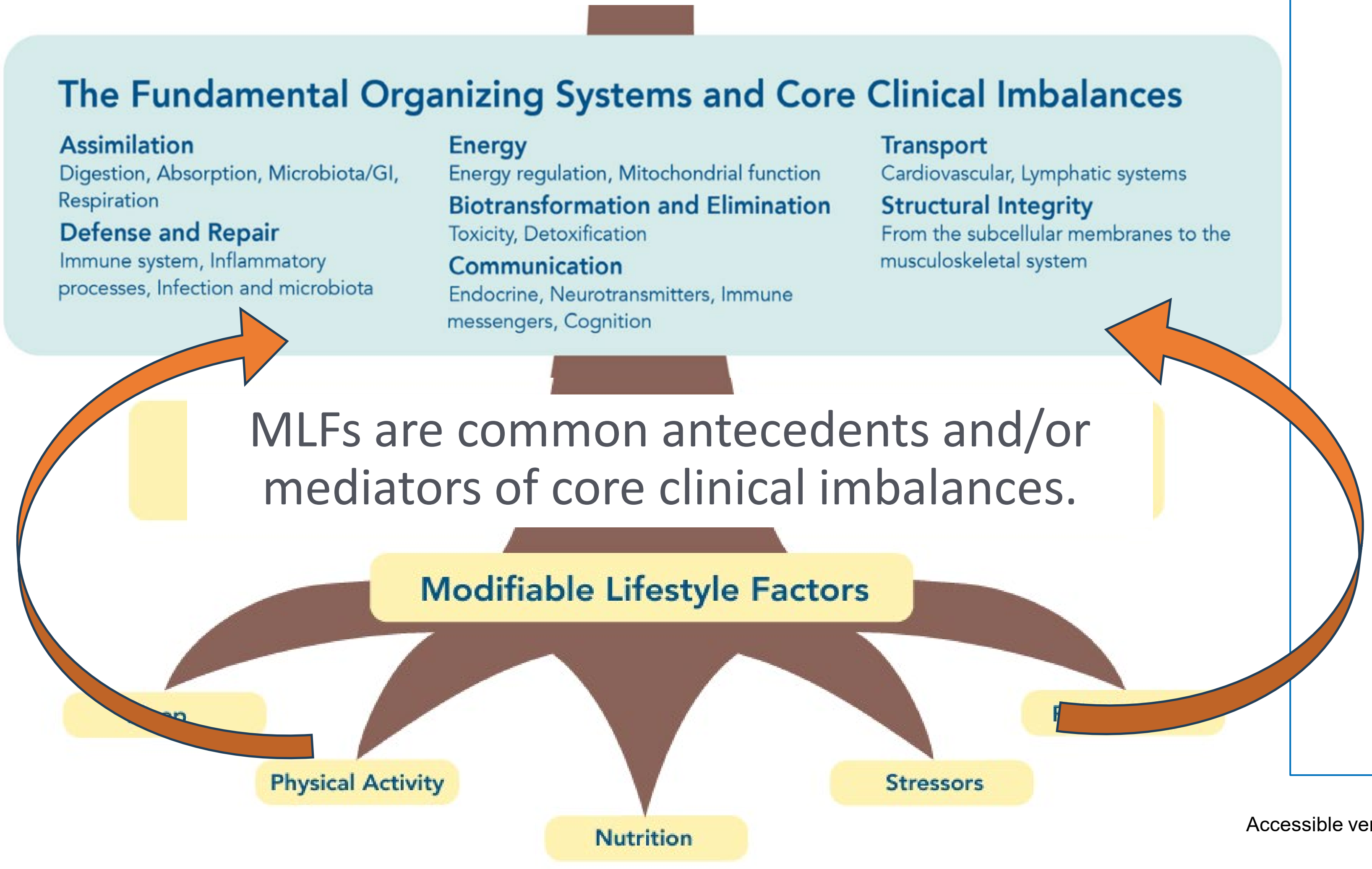
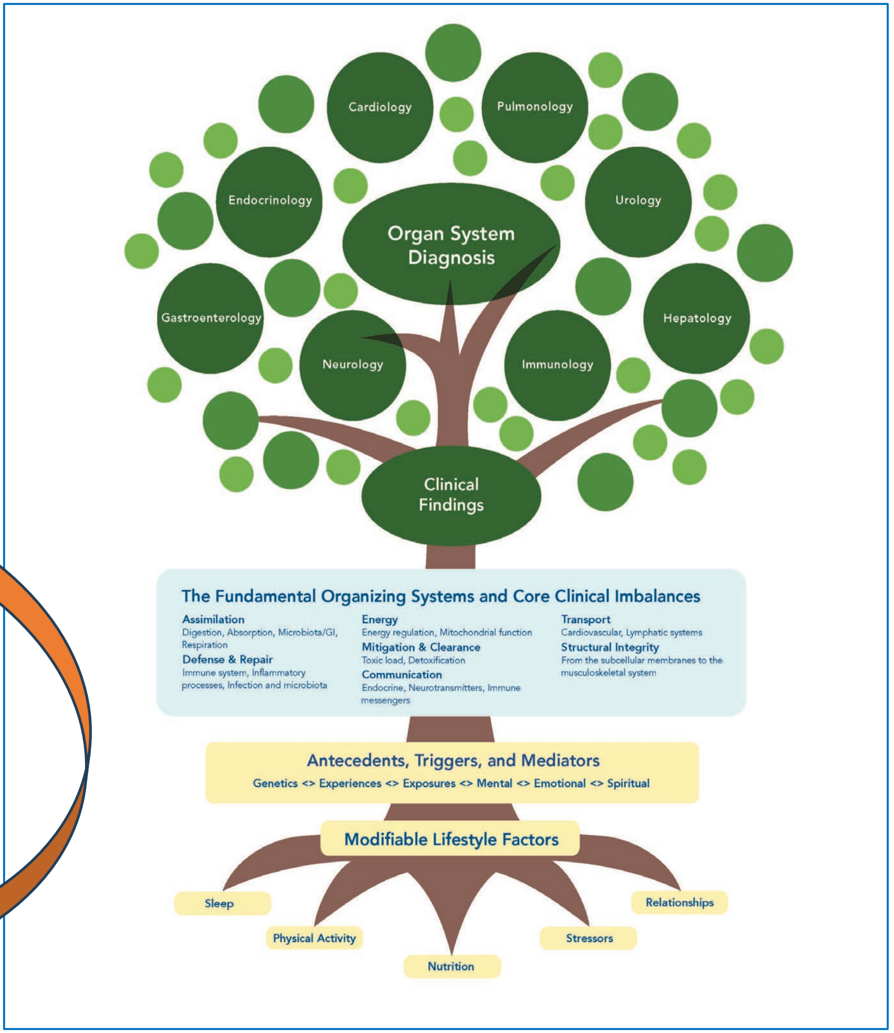
Lifestyle factors as root causes of physiological dysfunction and clinical imbalances.





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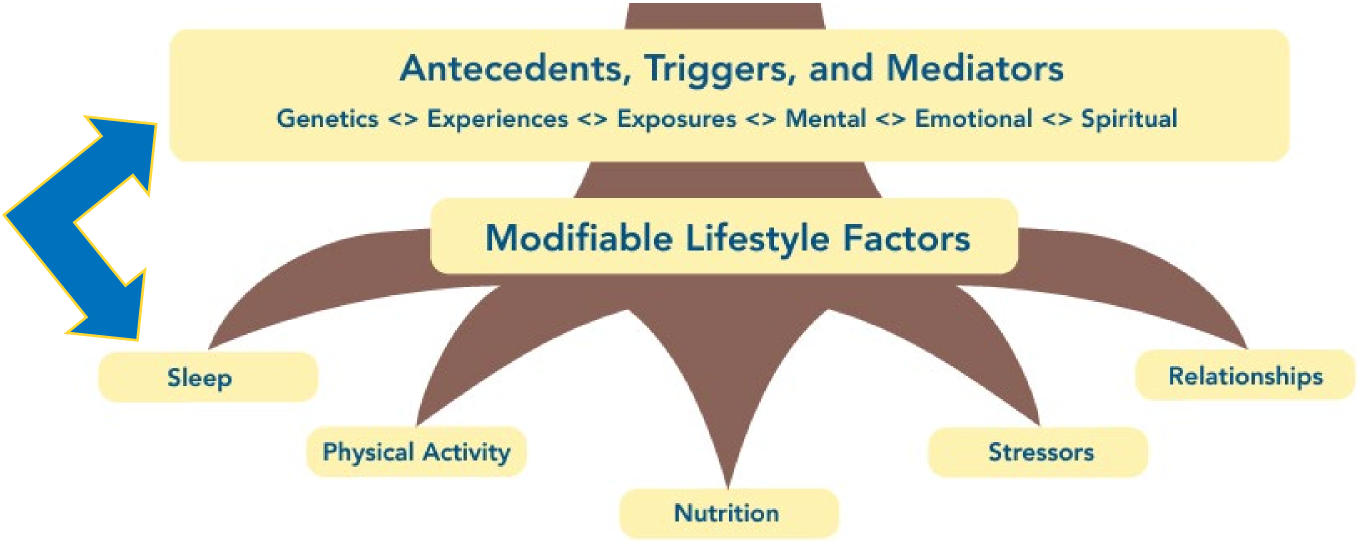
Lifestyle Factors as the Roots of the FM Model

The Role of MLFs in the Treatment of the Patient

MLFs may modulate genetic traits, as well as mental, emotional, and spiritual states.



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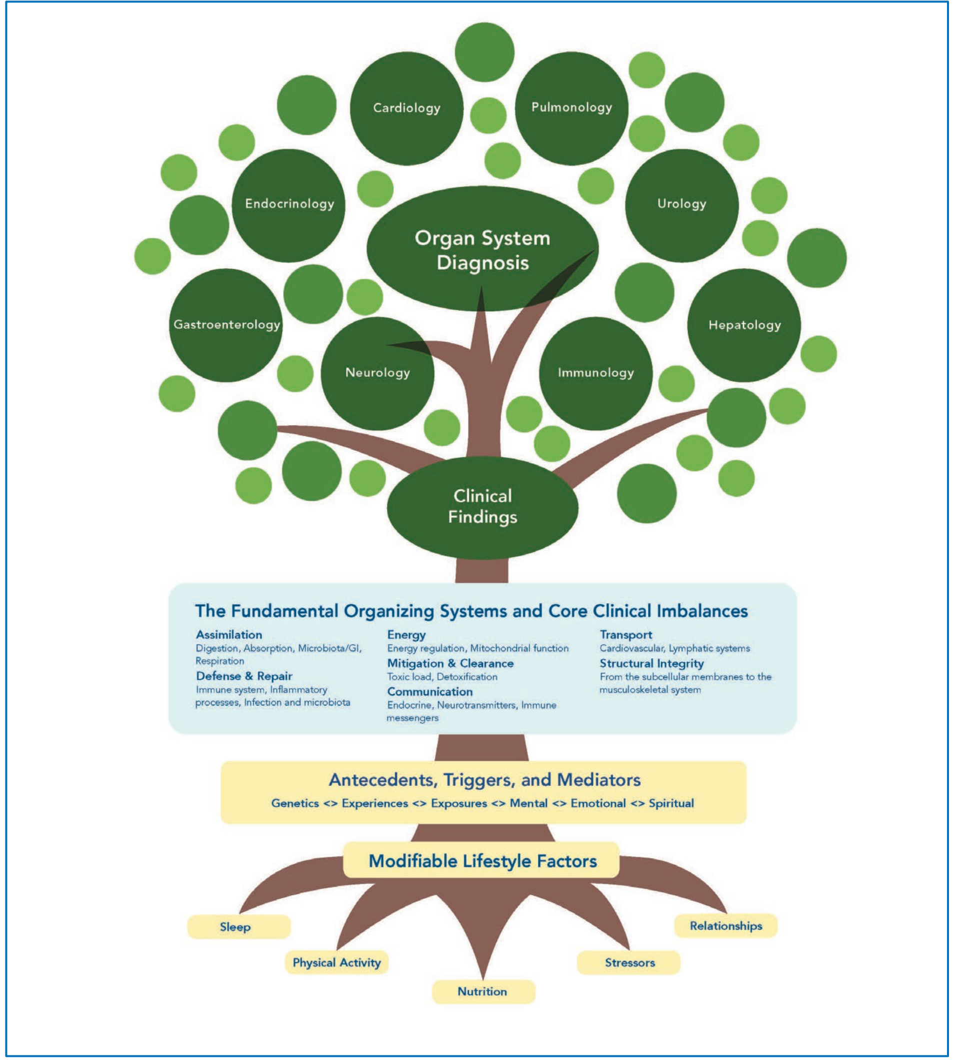

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Lifestyle Factors as the **Base** of the IFM Functional Medicine **Matrix**

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Functional Medicine Matrix (Teaching)

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The Patient's Story

Antecedents (Inherited & Acquired)

(Predispose or confer susceptibility; effects delayed by months to years)

- Inherited: FHx, genetics, maternal preconception, pregnancy environment
- Acquired: birth Hx, ACEs, SDOH, toxic exposure, smoking, malnutrition, physical inactivity, formula fed

Triggers & Precipitating Events

(Single, episodic or recurrent exposure, factor, or event; activates dysfunction within seconds to weeks)

- Examples: injury, trauma, procedure, biochemical exposures

Mediators/Perpetuators

(Current and ongoing; perpetuates symptoms, dysfunction, and/or disease)

- Examples: current dietary pattern, lifestyle, medication, toxic exposures, psychosocial factors

Clinical Imbalances and Physiological Dysfunctions

Assimilation
(e.g., digestion, absorption, microbiota/GI, respiration)

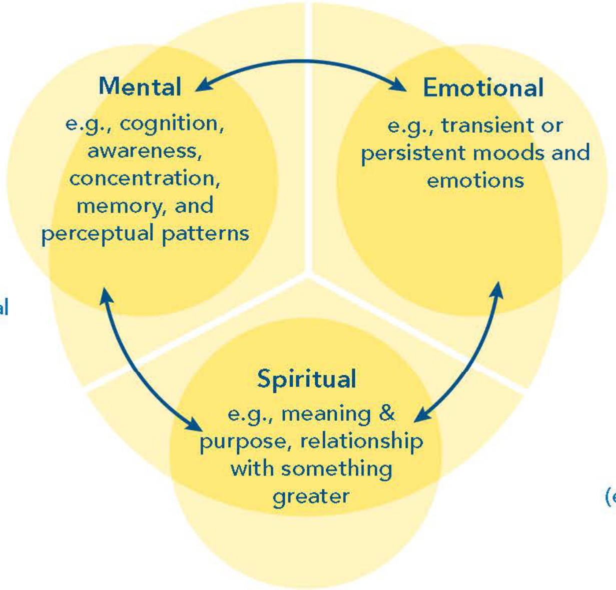
Defense & Repair
(e.g., immune, inflammation, infection/microbiota)

Structural Integrity
(e.g., from subcellular membranes to musculoskeletal structure)

Energy
(e.g., energy regulation, mitochondrial function)

Communication
(e.g., endocrine, neurotransmitters, immune messengers)

Mitigation & Clearance
(e.g., toxic load, detoxification)



Transport
(e.g., cardiovascular, lymphatic system)

Modifiable Lifestyle Factors (Focus on pertinent negatives for potential modification. Consider coping mechanisms related to MES)

Sleep

- Sleep quantity
- Sleep quality (efficiency, latency, depth)
- Sleep hygiene
- Sleep satisfaction
- Daytime sleepiness

Physical Activity

- Obstacles for movement: environment, pain, time, etc.
- Exercise frequency, intensity, time, and types (FITT)

Nutrition

- Dietary patterns
- Nutrient deficiencies or excess eating habits
- Food preferences & access
- Dietary allergies, intolerances, or sensitivities

Stressors

- Examples: financial, moving, work related stress, weather or environmental conditions, care giving, parental stress, divorce, loss of loved one

Relationships

- Community & support system: isolation, marginalization, lack of access to support networks
- Interpersonal relationships: lack of emotional support, high-conflict relationships

Abbreviations: ACEs = adverse childhood experiences; SDOH = social determinants of health; FHx = Family history, Hx = History

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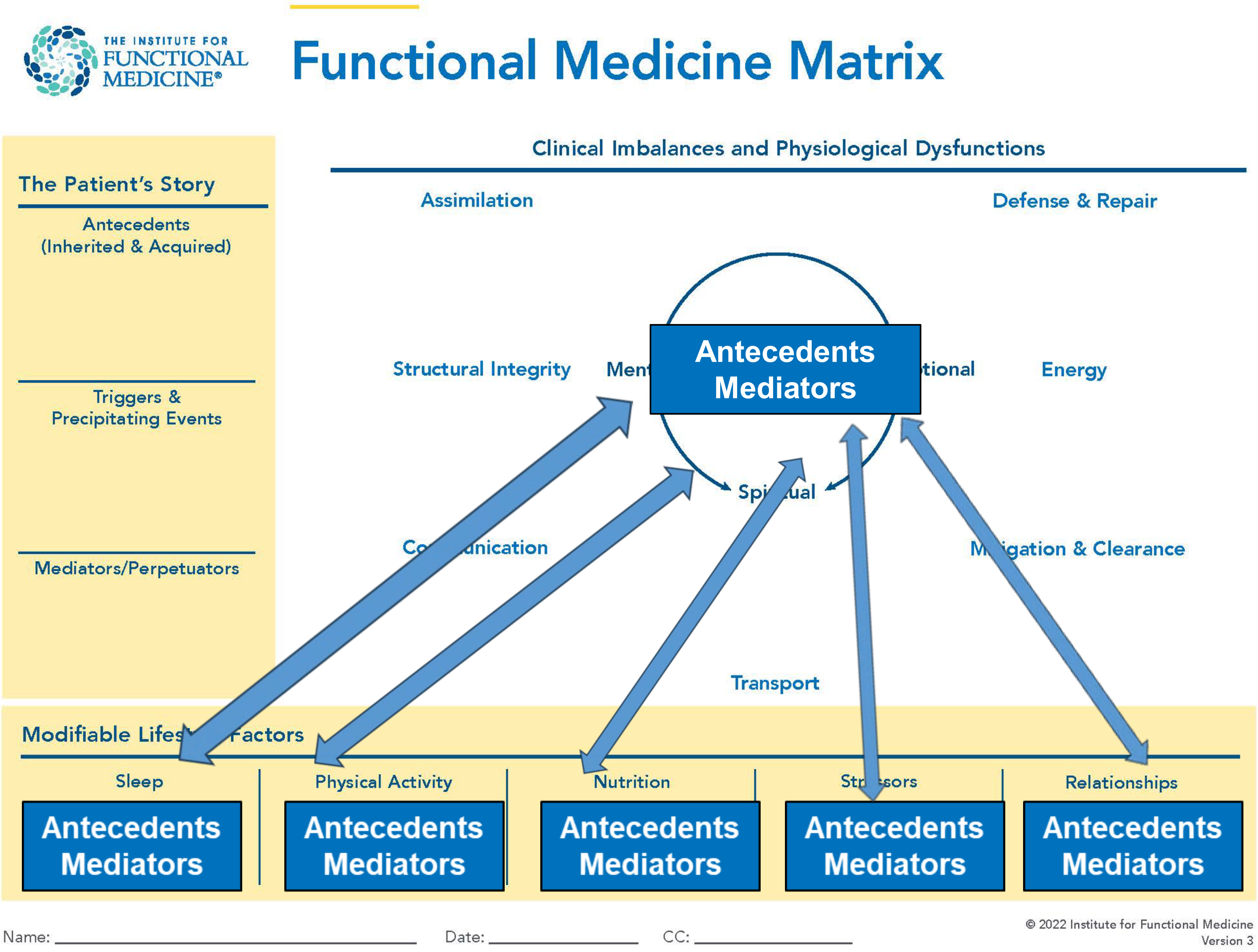
Bidirectional Effects on the Matrix

Modifiable lifestyle factors and mental, emotional, and spiritual factors may have bidirectional or mutually reinforcing effects on each other as antecedents and mediators of physiological dysfunction on the nodes of the matrix.



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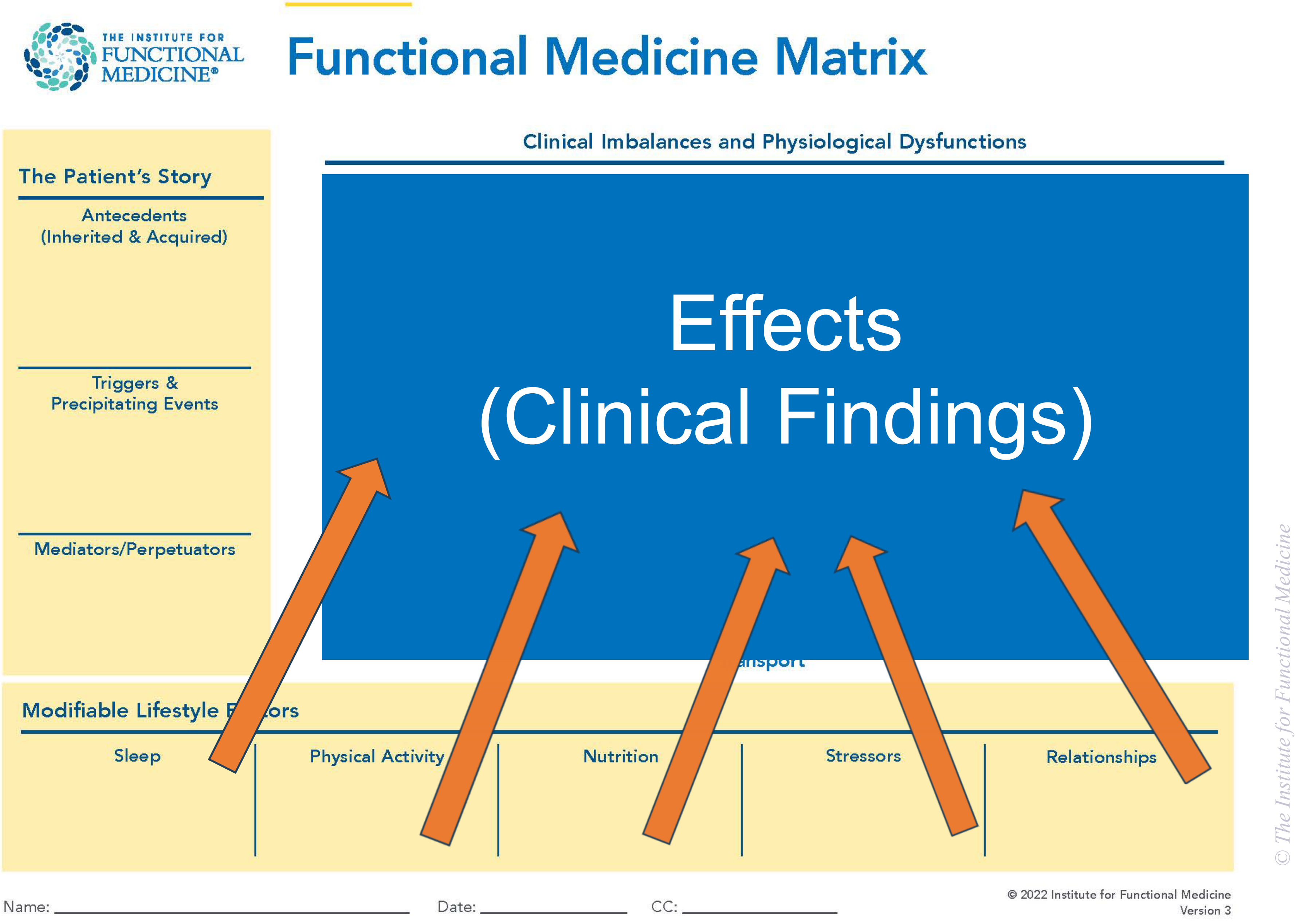
Modifiable lifestyle factors exert clinically significant mechanisms of action on physiological functions, thereby contributing to the effects of disease or the restoration of health.



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Modifiable Lifestyle Factors:

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